Inhibitory effect of oatmeal extract oligomer on vasoactive intestinal peptide-induced inflammation in surviving human skin.

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The aim of this study was to evaluate the antiinflammatory effect of oatmeal extract oligomer on skin fragments stimulated by a neuromediator, vasoactive intestinal peptide (VIP). Skin fragments (from plastic surgery) were maintained in survival conditions for 6 h. To induce inflammation, VIP was placed in contact with dermis by culture medium. Histological analysis was then performed on hematoxylin- and eosin-stained slides. Edema was evaluated with semiquantitative scores. Vasodilation was studied by quantifying the percentage of dilated vessels according to scores and by measuring their surface by morphometrical image analysis. TNF-alpha dosage was made on culture supernatants. Vasodilation was significantly increased after application of VIP. After treatment with oatmeal extract oligomer, the mean surface of dilated vessels and edema were significantly decreased compared with VIP-treated skin. Moreover, treatment with this extract decreased TNF-alpha.